

DPCOE INTERNAL HACKATHON

Organized by DPCOE Coders Club and Cyber Security Club

Problem Statement ID - SIH25131

Problem Statement Title - Student Innovation:
Swadeshi for Atmanirbhar Bharat - MedTech /
BioTech / HealthTech

Theme - MedTech / BioTech / HealthTech

PS Category - Software

Team ID - 53641

Team Mentors -
Prof. Vishakha Jadhav

Team Hexa.Coders

NAME	DEPARTMENT	CLASS
Apurva Ramkrushna Chavan	CSE	BE
Rohan Dipak Patil	CSE	BE
Vaishnavi Ganesh Bedwal	CSE	BE
Arti Vijay Nale	CSE	BE
Vrushali Sachin Bhandare	CSE	BE
Aniket Gokul Jagtap	CSE	BE

DOCUMENTATION

“Rakta Sanjivani”

Problem Statement:

India continues to face delays in arranging blood during emergencies due to the absence of a unified digital system that displays real-time availability across hospitals and blood banks. Most coordination still happens through manual phone calls and outdated records, causing confusion and slow response—especially when rare blood groups are urgently needed. Patients and families often panic as they contact multiple centers, while hospitals struggle with scattered or inaccurate stock data. This lack of coordination leads to shortages, wastage, and life-threatening delays.

Rakta Sanjivani addresses this gap by providing a real-time, location-based, and connected platform for hospitals, blood banks, donors, and patients to ensure faster, accurate, and reliable access to blood during critical situations.

Innovation:

1. Real-Time Blood Stock Visibility

Live, continuously updated blood availability across hospitals and blood banks—removing manual calls and delays.

2. Geo-Location–Based Blood Search

AI + GPS system that instantly shows the nearest facility with the required blood group, making emergency response faster.

3. AI-Driven Shortage Prediction

Machine learning models analyze trends to forecast upcoming shortages, helping hospitals prepare in advance.

4. Smart Notifications & Alerts

Automatic alerts to donors during urgent needs and timely notifications to blood banks for low stock, high demand, or SOS cases.

5. Unified Digital Network

A single connected platform linking hospitals, blood banks, donors, emergency services, and patients for smooth coordination.

6. Rare Blood Group & Disaster-Time Support

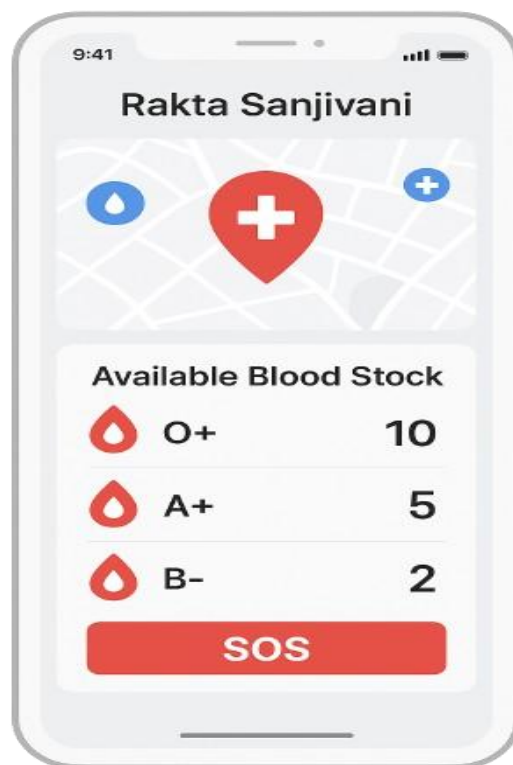
Special prioritization logic to quickly match rare blood groups and improve response during accidents or natural disasters.

7. Emergency SOS Feature

Immediate SOS trigger that notifies nearby donors and facilities, ensuring rapid support in critical situations.

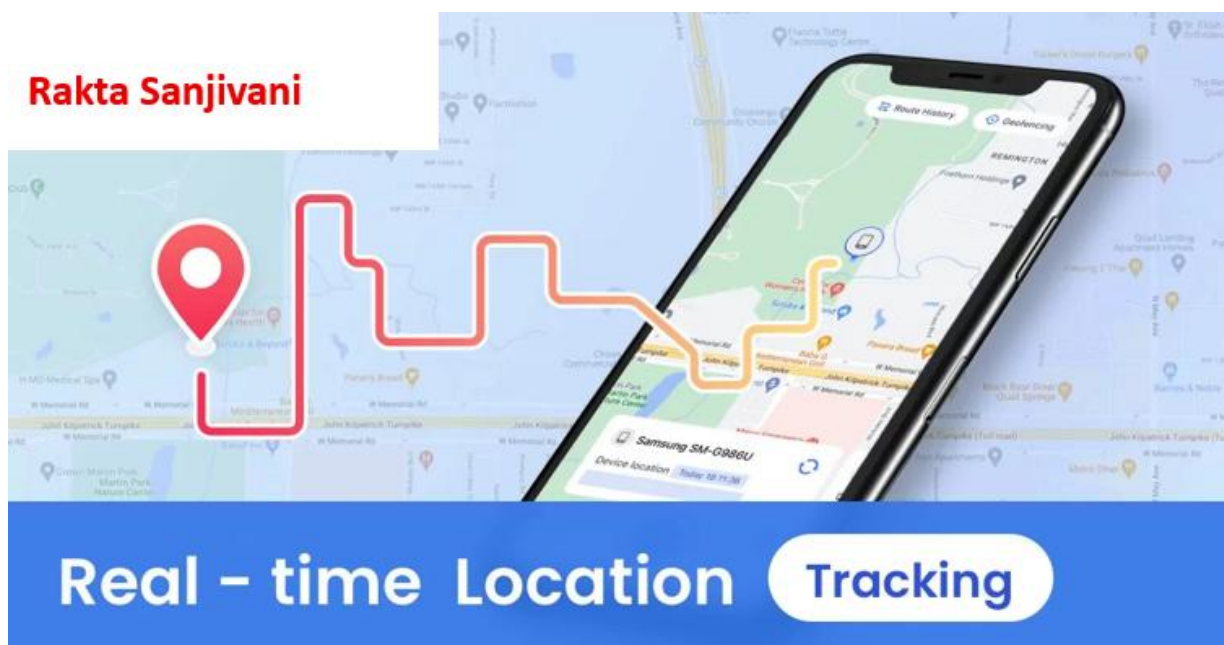
Objectives:

1. To provide real-time and accurate blood availability across hospitals and blood banks on a unified digital platform.
2. To reduce gaps and delays in emergency blood arrangements by replacing manual communication with instant digital updates.
3. To analyze and predict blood shortages using AI/ML for early planning and better preparedness.
4. To improve coordination between hospitals, blood banks, and donors for faster and more efficient emergency response.
5. To support rare blood group requirements using smart search, geolocation tracking, and priority alerts.
6. To send smart notifications to donors when their blood group is urgently needed or when donation intervals are due.
7. To provide instant alerts to blood banks and hospitals during shortages, high demand, or SOS requests.
8. To ensure quick, reliable, and user-friendly access for patients and families during critical situations through mobile and web platforms.



Problem Relevance:

- India continues to face frequent delays in arranging blood during emergencies because hospitals and blood banks lack a unified real-time system.
- Most facilities still depend on manual calls and scattered data, leading to slow response times—especially for rare blood groups.
- As the number of accidents, surgeries, pregnancy complications, and chronic illnesses increases, the demand for quick and reliable blood access becomes even more critical.
- With growing population needs, outdated communication methods, and poor coordination between hospitals, blood banks, and donors, there is an urgent requirement for a modern, digital, and data-driven solution.
- Rakta Sanjivani is highly relevant as it directly addresses these gaps by providing real-time availability, instant alerts, AI-based shortage prediction, and a unified nationwide network—making it essential for emergency care, disaster management, and overall healthcare improvement.



Feasibility:

1. Technical Feasibility

Rakta Sanjivani can be built using proven technologies such as GPS-based location services, real-time databases, APIs, cloud hosting, and AI/ML prediction models. Modern frameworks like React, Node.js, FastAPI, MongoDB, and AWS/GCP make the system scalable, secure, and capable of handling high traffic. All required technologies are widely available and easy for the team to integrate.

2. Operational Feasibility

Hospitals, blood banks, donors, and patients can use the platform with minimal training due to its simple dashboard and mobile interface. Real-time updates, smart alerts, and automated notifications ensure smooth day-to-day operations. The system can run 24/7 and supports both urban and rural usage.

3. Regulatory Feasibility

The solution aligns with India's blood safety guidelines and healthcare data rules. Secure authentication, encrypted storage, and controlled access ensure compliance with medical data privacy standards. Integration with authorized hospital/blood bank systems is permitted under existing regulations.

4. Economic Feasibility

The platform is cost-effective because it uses cloud infrastructure, open-source technologies, and reusable components. It reduces wastage of blood, lowers operational cost for hospitals, and improves efficiency, making it sustainable in the long run.

5. Team Feasibility

Your team has expertise in full-stack development, mobile apps, AI/ML, cloud services, and real-time systems. This makes the implementation realistic and achievable within the expected timeframe.

Requirements:

A. Functional Requirements

1. User Authentication & Roles
Separate login for hospitals, blood banks, donors, and patients.
2. Real-Time Blood Stock Dashboard
Live updates of available blood units across all centers.
3. GPS-Based Nearest Blood Center Locator
Shows the closest facility with the required blood group.
4. Emergency SOS System
Allows users to raise urgent requests and notify nearby centers/donors.
5. Smart Notifications to Donors
Alerts donors when their blood type is urgently needed.
6. Alerts to Hospitals/Blood Banks
Auto-alerts for shortages, high demand, or critical inventory levels.
7. AI-Based Shortage Prediction
Predicts future stock needs using historical and real-time data.
8. Communication Panel
Direct messaging/notification system between hospitals and blood banks.

B. Technical Requirements

1. Frontend: React / Next.js, typescript , javascript, Tailwind CSS
2. Backend: Node.js
3. Database: MySQL
4. AI/ML: Power BI,
5. APIs: Geolocation API, SMS

C. Client-Side Hardware

1. Smartphones (Android/iOS) for patients, donors, and hospital staff using the mobile app.
2. Laptops/Desktops for hospitals and blood banks to manage dashboards and update stock.

D. Networking & Connectivity

1. Stable Internet Connection for real-time data syncing.
2. Wi-Fi/4G/5G enabled devices for on-the-go updates and emergency alerts.

Applications

1. Hospitals

Quickly check real-time blood availability.

Improve treatment speed during emergencies.

Coordinate directly with nearby blood banks.

2. Blood Banks

Maintain updated stock records.

Receive alerts for shortages and high-demand cases.

Easily coordinate supply with hospitals.

3. Patients & Families

Instantly locate required blood units during critical situations.

Raise SOS alerts for emergency support.

Reduce panic and save valuable time.

4. Donors

Get smart notifications when their blood type is urgently needed.

Schedule donations and track eligibility.

Participate in life-saving efforts more effectively.

5. Government & Health Authorities

Monitor national and regional blood availability.

Improve decision-making during disasters or crises.

Support public health programs and donation drives.

6. Rural & Remote Areas

Access nearby city hospitals' blood stock when local options are limited.

Improve emergency preparedness and response times.

